Do Students Agree with LLM Feedback on Their Own Code? A Within-Subject Study of Four Feedback Configurations in Software Project Courses

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Abstract. LLM-generated feedback can support comprehension checks on students' own software projects, but perceived quality and willingness to use the feedback need not coincide. We evaluated PACE, a repository-grounded question-and-feedback system, with 86 students from 28 team repositories. Each student answered four questions and rated one instance of each of four feedback configurations assigned by random permutation: Sandwich, Rubric, Explainable and Researching. The configurations differ in content structure and, for Researching, the intended model pipeline. Across 344 instances, an exploratory fourteen-item composite averaged 4.07/5 (95% repository-cluster bootstrap CI [3.90, 4.23]); 79.9% of ratings endorsed the AI's judgement, 68.0% endorsed future use and 41.9% endorsed motivation. Configuration differences in the composite were small (exploratory Friedman $\chi^2(3) = 23.04$, $p < .001$, Kendall's $W = .089$). The configurations designed to include explicit next steps had higher mean actionability, but the design does not isolate the effect of next steps. All six repository-bootstrap 90% intervals for composite mean differences lay within an exploratory ±0.25-point margin; Acceptance-item intervals included zero. An observational within-student comparison of raw-code and graph contexts was inconclusive ($b = 0.07$, 95% CI $[-0.02, 0.20]$). These findings concern students' perceptions, rather than expert-verified correctness, actual revision or learning gains.

Keywords: LLM-generated feedback · Feedback acceptance · Programming education · Software engineering education · Human–AI trust

1 Introduction

Feedback is one of the strongest levers on learning, but its effect depends on what the learner does with it [12,31]. Technically correct feedback does not by itself demonstrate that students will understand or use it. Research on feedback literacy therefore treats the student's uptake—making sense of a comment, judging it, and deciding to act—as part of the feedback process rather than its aftermath [2,34].

LLMs can identify issues in student code and explain them in readable language, but their comments may also be misleading [14,20,4]. Guardrails can support guidance without supplying complete solutions [25]. Their growing role in computing education [6,17]

makes student acceptance worth examining separately from technical quality. Students valued code-specific feedback in a programming study [35]; an ENL writing study found split preferences between tutor and AI feedback [10]. A Java-course experiment favoured instructor feedback [9], while provenance disclosure can alter perceptions [27].

Two gaps motivate this study. First, project courses assess evolving team repositories that no student wrote alone. Questions about learners' own programs can probe comprehension [24], but perceptions of LLM feedback on those answers remain underexplored in the literature discussed here. Second, deployed systems bundle praise, criteria, evidence and model orchestration into different configurations, each with a pedagogical rationale [28,16,19,8]. We compare students' ratings of these bundles without assuming identical underlying judgements.

We address these gaps with PACE (Project Assistance and Course Evaluation), a platform deployed in software project courses at FPT University. PACE analyses each team's repository, generates scenario questions about business rules implemented in it, and returns LLM feedback on every answer. Each student answered four questions and received feedback in four different configurations, assigned by random permutation, then rated every feedback instance on fifteen items covering perceived quality, utility, trust, affect, and acceptance. We ask:

RQ1 How do students judge the quality of LLM feedback on their answers about their own code, and how far do they accept it?
RQ2 Does student agreement differ across the four deployed feedback configurations?
RQ3 How is student agreement associated with raw-code versus graph-based question contexts in the deployed system?

Our contributions are a repository-grounded comprehension workflow, a field evaluation with 86 students and 344 rated instances, and an analysis separating perceived quality, acceptance and motivation. We compare complete configurations, examine response-pattern sensitivity and report uncertainty in an observational context comparison.

2 Related Work

2.1 LLM-Generated Feedback in Programming Education

Automated programming feedback predates LLMs: a review of 101 tools found more emphasis on identifying mistakes than on supporting next steps [18]. Hellas et al. found that GPT-3.5 often identified an actual issue in novice programmers' requests but also missed issues or invented them [14]. Kiesler et al. similarly cautioned that formative programming feedback could mislead novices [20]. Dai et al. compared ChatGPT and instructor feedback [4], while CodeHelp demonstrated guarded assistance at course scale [25]. We complement these evaluations with student perceptions of feedback on comprehension answers about team repositories, extending questions about learners' own code [24].

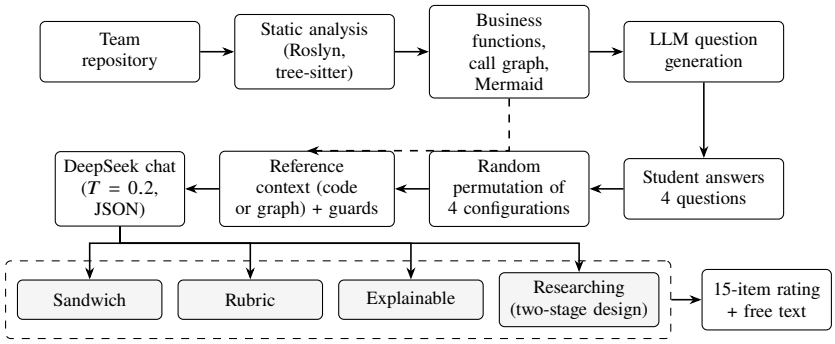


Fig. 1. Overview of PACE. Questions and the evaluator's reference context are both derived from the student's own repository. Each student's four answers receive the four configurations in a random order.

2.2 Student Perceptions and Acceptance of AI Feedback

Usefulness and trust are related to intentions to continue using a system [5,33], while appropriate reliance requires calibration to actual reliability [23]. Algorithm aversion [7] and appreciation [26] both occur. In programming education, Zhang et al. reported preferences for code-referencing feedback [35]; Er et al. found advantages for instructor feedback [9]; and Nazaretsky et al. found perceptions sensitive to AI provenance [27]. These findings motivate measuring trust and intention alongside perceived quality.

2.3 Feedback Configurations

Sandwich feedback improved perceptions without improving performance in a controlled study [28]. Rubrics make criteria explicit [16]. Evidence-based explanations may support appropriate trust [19]. Multiagent debate has improved factuality on benchmark tasks [8], but our sequential inspection–synthesis pipeline does not implement debate. Model-as-judge benchmarks also document position and verbosity biases alongside agreement with human preferences [36]. Neither result validates PACE feedback against expert judgements. Our within-student comparison evaluates complete bundles.

3 The PACE System

Figure 1 summarises the pipeline. PACE consists of an ASP.NET Core (.NET 9) back end, a PostgreSQL database and a Next.js front end; the parts relevant to this study are repository analysis, question generation, and feedback generation.

3.1 Repository Analysis and Question Generation

For each team repository, PACE clones the submitted branch and parses it with Roslyn (C#) or tree-sitter (Java, JavaScript and TypeScript). Endpoints are grouped into *business functions* (e.g., "contract management" or "payment confirmation"), either from

a template stored in the repository or from a draft that an instructor reviews and confirms. For every function, PACE resolves calls from each entry point and stores both the source code of the involved methods and a data-flow graph with ordered call edges, branch conditions and a Mermaid rendering.

Questions are generated per business function by the configured question-generation LLM (temperature 0.3, structured JSON output), which receives either the raw source code or the Mermaid graph; a router selects the graph when the flow involves more than ten distinct methods and the raw code otherwise. The prompt targets scenario questions about guard conditions and failure branches, for example: *"When the facility manager tries to swap the storage unit of a contract that still has an outstanding balance, how does the system respond and why is the action blocked?"* Each question is stored with a suggested answer and with the context type used to produce it.

3.2 Feedback Generation

When a student submits answers, each answer is evaluated against a *reference context* built from the same business function: the raw source of its methods, or its flow graph with numbered steps and conditions, following the context type of the question. The system prompt fixes an instruction hierarchy: content inside the `<student_answer>` tag is data, never instructions; the reference context and the suggested answer supply the evaluation reference, whose correctness is not independently established here; the model comments and explains but does not assign an overall grade; and the output must be a single JSON object. An input guardrail rejects answers that contain prompt-injection phrases or executable payloads, and an output guardrail validates the JSON before it is stored.

The four configurations bundle different content requirements and generation procedures:

- **Sandwich**: positives, constructive criticism, and next steps.
- **Rubric**: a list of criteria that the model names for the question (e.g., understanding of the business flow, identification of the guard condition), each with a 1–5 score and a comment.
- **Explainable**: a *code evidence* field that cites methods, steps or conditions from the context, and a *logic explanation* that relates them to the answer.
- **Researching**: a two-stage design: a DeepSeek model checks the answer against the code, then GPT-OSS-120B (served by Groq) synthesises a report: an overall comment, positives, criticism, next steps, code evidence, two scored criteria, and a consensus summary with pedagogical guidance.

Only Sandwich and Researching are designed to require an explicit list of next steps. The three single-model configurations specify `deepseek-chat` (temperature 0.2, JSON mode). This API alias does not identify an immutable model version. If the second model in the Researching pipeline fails, the system falls back to the single-model path; all 86 Researching records list both roles in nested metadata (DeepSeek-V3 and Groq-GPT-OSS-120B). This supports the recorded two-model configuration; independent provider execution logs are not part of the export.

Table 1. Survey instrument (translated from Vietnamese). †Reverse-worded.

Construct	Item	Statement
Perceived quality	Accuracy	The AI's comment correctly identifies the problems in my answer.
	Specificity	The AI points to exactly which part of my answer needs revisiting, rather than commenting in general.
	Clarity	The wording and presentation of the comment are clear and unambiguous.
	Transparency	I understand which academic criteria the AI used to assess my answer.
Perceived utility	Usefulness	This AI tool is an effective learning assistant for my project work.
	Actionability	I know exactly which technical steps to take to improve my answer.
Trust	Competence	I trust the technical expertise and analytical capability of this AI.
	Reliability	The AI assesses consistently and does not change its standards at random.
	Fairness	The AI's assessment fairly reflects the effort and actual quality of my answer.
Affect	Affect	I feel motivated to keep learning after reading the AI's comments.
	Psych. safety	AI feedback causes less pressure than being criticised by a person.
Acceptance & intention	Acceptance	I agree with the AI's judgement of my answer's strengths and weaknesses.
	Willingness	Based on this comment, I am ready to revise my answer right away.
	Behav. intention	I intend to keep using this LLM system for feedback in future projects.
Separate item	Rejection†	I tend to want to refute or argue against the AI's assessment.

The assigned configuration is the experimental manipulation. Comparisons use assigned labels, including any system guardrail responses. When a student submits $n = 4$ answers, PACE shuffles the four configurations and assigns one to each answer, so every student sees every configuration exactly once, in random order.

4 Study Design

4.1 Context and Participants

The study took place in October 2026 in undergraduate software engineering project courses at FPT University, Vietnam. The research team visited classes and invited students to use PACE and provide candid feedback as preparation for questions they might encounter before a project defence panel. Students used the system to rehearse explaining their team's repository. Recruitment was classroom-based rather than a probability sample. The sample comprises 86 students from 28 team repositories, answering 190 distinct questions on 152 business functions. Answers were written in Vietnamese and were short, about 51 whitespace-separated tokens on average. Submissions were made between 20 and 26 October 2026.

4.2 Instrument

After reading each feedback instance, students rated it on fifteen statements on a five-point scale (1 = strongly disagree, 5 = strongly agree) and could add a free-text reply. The items were written in Vietnamese by the research team, drawing on formative-feedback principles [31] and technology-acceptance constructs [5,33,23]. Table 1 lists them with English translations, grouped into five a priori constructs. Rejection is reverse-worded.

4.3 Data Set

The survey export contained 344 answer records from 86 submissions, one submission per student. We screened every record on metadata only (no explicit development-fixture marker in the export, feedback stored, all fifteen ratings present); all 344 passed, so none was excluded. Every student rated all four configurations, which gives 86 complete blocks and 86 instances per configuration. One assigned Rubric instance contains a guardrail refusal with next steps but no criteria; the other 85 contain criteria. We retain assigned labels rather than reclassifying outputs. The export also stores the context type of each question: 280 instances (157 questions) came from graph-based questions and 64 instances (33 questions, 13 repositories) from raw-code questions. Thirty-two students received at least one raw-code question, and 26 of them answered questions of both types.

4.4 Analysis

Scores. The *AI agreement index* (AAI) is an exploratory composite: the mean of fourteen non-reversed items across several domains. It is not a validated unidimensional measure of agreement. Construct scores are item means, and the Acceptance item is also reported separately. The decision to exclude Rejection followed inspection of its inconsistent behaviour (Section 5.1); it is therefore a data-informed scoring choice [32].

Reliability and data quality. We report Cronbach's α [3] and McDonald's ω_t from a one-factor solution [13], and flag *straight-lined* responses—identical ratings on all fifteen items—following Kim et al. [21].

RQ1. We report item means, distributions and 95% percentile confidence intervals from resampling all 28 repositories with replacement (5,000 replicates; seed 20261029). Each sampled repository contributes all its students and questions; means weight feedback instances equally. These intervals allow dependence within repositories, assuming independence across repositories [1]. To explore conditional associations with acceptance, we fit student fixed-effects regressions after demeaning predictors and outcomes within each student:

$$y_{ij} - \bar{y}_j = \sum_{k=1}^{5} \beta_k (z_{kij} - \bar{z}_{kj}) + \sum_{c=2}^{4} \gamma_c (D_{cij} - \bar{D}_{cj}) + e_{ij}, \tag{1}$$

Here i indexes feedback instances, j students, y is Acceptance, Willingness or Behavioural intention, and D_c are configuration indicators (Sandwich reference). Predictors are perceived quality, perceived utility, trust, motivation and psychological safety, standardised using their full-sample SDs before demeaning. We separate the two affect items because their joint reliability is low. Regression intervals use the same repository bootstrap. These models describe contemporaneous within-student associations, not causal drivers or differences between students.

RQ2. We describe mean differences using repository-bootstrap intervals and use a Friedman test [11] on AAI over the 86 complete blocks, with Kendall's W as effect size. We apply separate Holm corrections across five construct tests, fifteen item tests, and six paired Wilcoxon tests [15]. Rank-test p-values assume independent student blocks and are exploratory because repositories are shared. Because the Friedman test uses only the within-student rank order, we also fit a linear mixed model, AAI $\sim$ configuration $+ (1 \mid \text{student})$, compared to an intercept-only model by a likelihood-ratio test, and a second model that adds feedback length and answer length as covariates. Because significance says nothing about whether a difference matters, we examine an equivalence criterion [22]: the repository-bootstrap 90% interval of each paired mean difference must lie within ± 0.25 points, a quarter of one scale step. This pragmatic, exploratory margin has not been calibrated against learning outcomes or elicited from stakeholders; equivalence claims are restricted to AAI under this margin. Sensitivity analyses retain complete student blocks after excluding straight-lined rows or the guardrail-refusal block, and recompute contrasts using all fifteen items with Rejection reverse-scored.

RQ3. Context type was not randomised: the router uses graph context for functions involving more than ten methods and raw code otherwise. Complexity, question context and evaluator context are therefore coupled. We report descriptive means and an exploratory student fixed-effects model with context and configuration indicators, using repository-bootstrap intervals. The context coefficient uses within-student variation, supplied by 26 students exposed to both types. It remains an association confounded by complexity and is not an isolated context effect.

Dependence and scope. Answers are assigned within students, who share repositories and questions. Repository resampling preserves students and shared questions: each belongs to one repository. Rank tests and student-intercept mixed models do not fully capture repository dependence; substantive comparisons therefore emphasise repository-bootstrap intervals. Straight-lining is a response-pattern flag, not proof of inattention.

Analyses use Python and statsmodels [30]. Supplementary scripts accompany the paper. Identifier-bearing raw exports are withheld; an anonymised release remains to be prepared.

5 Results

5.1 Reliability and Data Quality

The fourteen non-reversed items had high internal consistency ($\alpha = .936$, $\omega_t = .939$). Item-group estimates were $\alpha = .91$ for perceived quality, .78 for utility, .80 for trust and .78 for acceptance & intention. Affect had lower consistency (.56): its two items, motivation and psychological safety, should not be assumed to form a single reliable construct. Internal consistency alone does not establish dimensionality or construct validity. Two observations qualify this.

First, 64 of the 344 responses (18.6%) were straight-lined, and 11 of the 86 students (12.8%) straight-lined all four of their ratings. Excluding straight-lined response rows lowers α only to .90, so the scale's consistency does not rest on straight-lining.

Table 2. Item ratings across 344 feedback instances. CI: 95% repository-bootstrap interval of the mean (28 repositories). Agree/Disagree: % of ratings ≥ 4 / ≤ 2. †Reverse-worded, not reverse-scored.

Item	Mean (SD)	95% CI	Agree	Disagree
Accuracy	4.38 (0.96)	[4.14, 4.61]	83.4	5.5
Specificity	4.36 (0.94)	[4.10, 4.60]	83.4	5.2
Clarity	4.33 (0.88)	[4.09, 4.56]	81.7	3.5
Transparency	4.29 (0.95)	[4.03, 4.54]	79.9	4.9
Usefulness	4.09 (0.77)	[3.93, 4.24]	84.0	2.9
Actionability	4.13 (0.88)	[3.90, 4.34]	80.8	4.9
Competence trust	4.17 (0.98)	[3.94, 4.41]	76.5	4.9
Reliability	3.88 (0.80)	[3.73, 4.01]	78.5	4.9
Fairness	4.15 (0.79)	[3.98, 4.31]	81.4	2.3
Affect	3.53 (0.87)	[3.38, 3.71]	41.9	4.9
Psych. safety	3.83 (0.85)	[3.65, 3.99]	76.2	7.3
Acceptance	4.09 (0.82)	[3.90, 4.27]	79.9	3.5
Rejection†	2.87 (1.22)	[2.56, 3.20]	30.8	48.8
Willingness	3.97 (0.69)	[3.86, 4.08]	81.4	2.3
Behav. intention	3.80 (0.85)	[3.65, 3.94]	68.0	5.2

Second, the association involving the reverse-worded REJECTION item differed with response-pattern exclusions. Across all responses, REJECTION correlated only weakly and negatively with ACCEPTANCE (Spearman $\rho = -.14$), and its reverse-scored item–rest correlation was close to zero ($-.03$). Once straight-lined responses were removed, the negative association was larger ($\rho = -.40$ with ACCEPTANCE and $\rho = -.41$ with the agreement index, descriptively). A student who rates every statement 4 also "agrees" to wanting to argue with the AI, a known weakness of reverse wording [32]; we therefore keep the item out of the exploratory composite. These correlations describe response patterns; row-level significance tests would ignore repeated ratings.

5.2 RQ1: How Students Judge the Feedback

Table 2 and Fig. 2 show that ratings were positive on every non-reversed item. The agreement index averaged 4.07 (SD 0.64, 95% repository-bootstrap CI [3.90, 4.23]). Students most often endorsed usefulness (84.0% agree), accuracy and specificity (83.4% each), and explicit disagreement stayed at or below 7.3% on all fourteen items. On the reverse-worded item, 48.8% rejected the idea of arguing with the AI and 30.8% endorsed it.

The profile is far from flat. Students readily said that a comment was correct, specific and clear, and most also extended trust to the system's competence (76.5%) and consistency (78.5%). They were less ready to commit to it: 68.0% intended to keep using PACE in future projects. The weakest item by a wide margin was affect: only 41.9% said the feedback made them motivated to keep learning, and more than half (53.2%) chose the neutral option. Endorsement of perceived quality was more frequent than endorsement of motivation.

Conditional associations with acceptance. Table 3 reports within-student coefficients and repository-bootstrap intervals. Utility was positively associated with acceptance and future-use intention. Trust was positively associated with acceptance. Perceived-quality

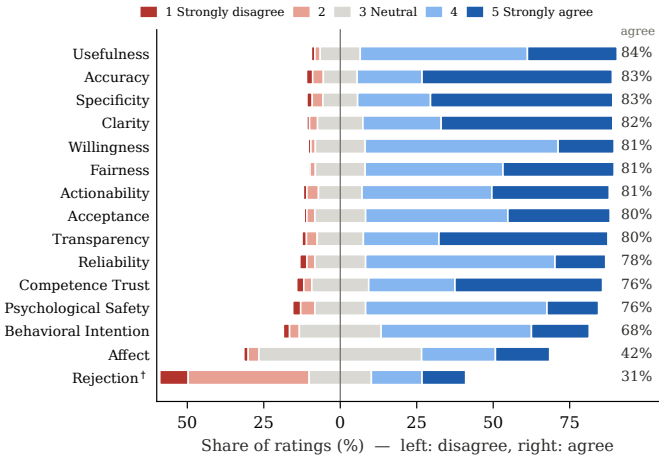


Fig. 2. Distribution of ratings per item, sorted by the share of agreement (right-hand labels). Neutral ratings are centred on zero.

Table 3. Student fixed-effects regressions (Eq. 1). Cells show rating points per full-sample predictor SD [95% repository-bootstrap CI]; 5,000 replicates. Motivation and psychological safety are separate predictors; configuration indicators are included.

Predictor	Acceptance	Willingness	Behav. intention
Perceived quality	0.19 [−0.03, 0.41]	0.03 [−0.23, 0.32]	0.10 [−0.14, 0.36]
Perceived utility	0.36 [0.20, 0.51]	0.14 [−0.06, 0.25]	0.21 [0.06, 0.35]
Trust	0.24 [0.09, 0.42]	0.06 [−0.08, 0.22]	0.13 [−0.14, 0.40]
Motivation	−0.09 [−0.20, 0.03]	0.27 [0.03, 0.46]	−0.05 [−0.20, 0.10]
Psych. safety	−0.05 [−0.12, 0.00]	−0.03 [−0.17, 0.07]	0.07 [−0.09, 0.25]

intervals included zero after configuration adjustment; without that adjustment its acceptance interval was [0.02, 0.44]. Motivation was positively associated with willingness; psychological-safety intervals included zero throughout. The fifteen exploratory intervals are not adjusted for multiplicity, and coefficient differences were not directly tested.

5.3 RQ2: Differences Between Feedback Configurations

The four configurations received similar but not identical ratings (Table 4, Fig. 3). Mean AAI ranged from 4.01 (Rubric) to 4.12 (Sandwich). The Friedman test detected a difference ($\chi^2(3) = 23.04$, $p < .001$), but its size was small ($W = .09$). Exploratory pairwise rank tests located it between Sandwich and the two configurations not designed to require next steps: Sandwich was rated 0.11 points above Rubric ($p_{\text{Holm}} = .002$) and 0.06 points above Explainable ($p_{\text{Holm}} = .023$); no other pair differed after correction. The mixed-model omnibus test was less conclusive (likelihood-ratio $\chi^2(3) = 6.65$, $p = .084$). Rank and mean-based analyses target different aspects of responses; the rank result does not establish a large mean difference.

Table 4. Ratings by feedback configuration, mean (SD). Friedman tests over the 86 students, each of whom rated all four configurations; p for the construct rows is Holm-adjusted. Each configuration has 86 instances.

	Sandwich	Rubric	Explainable	Researching	$\chi^2(3)$	p	W
Agreement index (AAI)	4.12 (0.69)	4.01 (0.66)	4.07 (0.61)	4.09 (0.60)	23.04	<.001	0.089
Perceived quality	4.36 (0.82)	4.33 (0.82)	4.39 (0.79)	4.28 (0.86)	6.31	.195	0.024
Perceived utility	4.27 (0.86)	4.01 (0.69)	3.95 (0.66)	4.22 (0.73)	45.57	<.001	0.177
Trust	4.12 (0.78)	3.91 (0.75)	4.11 (0.70)	4.12 (0.65)	24.98	<.001	0.097
Affect	3.74 (0.69)	3.65 (0.75)	3.63 (0.76)	3.72 (0.68)	10.22	.050	0.040
Acceptance & intention	3.95 (0.70)	3.91 (0.71)	3.96 (0.62)	3.98 (0.62)	3.70	.295	0.014
Feedback length (units)	248	267	215	623	172.19	<.001	0.667

The exploratory rank differences were concentrated in utility and trust. Perceived utility varied most ($\chi^2(3) = 45.57$, $p_{\text{Holm}} < .001$, $W = .18$): Sandwich (4.27) and Researching (4.22) were rated more useful and actionable than Rubric (4.01) and Explainable (3.95). At the item level this gap was largest for actionability (4.33 and 4.28 vs. 3.97 and 3.95; $p_{\text{Holm}} < .001$), with a Sandwich–Explainable mean difference of 0.37 (95% repository-bootstrap CI [0.16, 0.57]). The configurations differ in more than next steps, so this contrast does not identify that feature's effect. Trust also varied ($\chi^2(3) = 24.98$, $p_{\text{Holm}} < .001$, $W = .10$), with Rubric lowest (3.91 vs. 4.11–4.12); Rubric feedback also received the lowest ratings for competence trust (3.97) and fairness (3.95). Sandwich feedback was rated highest on the motivation item (3.69 vs. 3.41 for Explainable; item $p_{\text{Holm}} = .016$). No difference was detected for perceived quality or acceptance & intention, and no difference was detected on the Acceptance item across configurations (4.05–4.12, $p_{\text{Holm}} = 1.00$). All six 95% repository-bootstrap Acceptance contrasts included zero; for example, Rubric–Researching was -0.07 $[-0.27, 0.12]$. This does not establish item-level equivalence. Researching feedback had a lower mean clarity rating than the others (4.19 vs. 4.36–4.41), but this did not survive correction ($p_{\text{Holm}} = .22$).

Researching feedback was longer: its mean length was 623 whitespace-separated text units versus 215–267 for the other configurations ($\chi^2(3) = 172.2$, $p < .001$, $W = .67$). Adding feedback length and answer length to the mixed model left the estimated Rubric–Sandwich gap near -0.11 points ($p = .018$). The length-covariate intervals included zero. Feedback length is generated by the assigned configuration, so this adjustment does not identify a direct configuration effect or rule out a role for length. The length proxy counts strings in stored JSON, excluding identifiers, configuration labels and model names; it is neither Vietnamese lexical word count nor provider token usage. Cost comparisons require usage logs.

All six repository-bootstrap 90% intervals for paired AAI mean differences were inside the exploratory ±0.25 margin; Sandwich–Rubric was $[0.008, 0.212]$. This supports only small average composite differences under that chosen margin, not equal item profiles or a validated practical-equivalence threshold. Retaining all fifteen items with Rejection reverse-scored gave a mean of 4.01 and again placed all six intervals inside the margin. The Friedman result persisted after excluding straight-lined rows (64 complete students, $\chi^2(3) = 23.51$, $W = .12$) and after excluding the single guardrail-refusal

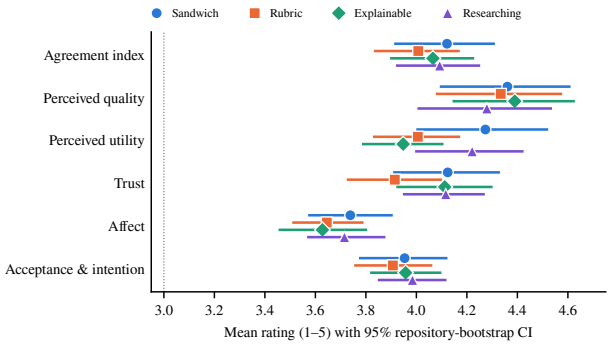


Fig. 3. Agreement index and construct scores by feedback configuration, with 95% repository-bootstrap confidence intervals (28 clusters). The dotted line marks the neutral midpoint.

Table 5. Agreement by question context type, mean. b: within-student estimate for raw code vs. graph, controlling for configuration.

Context	n	AAI	Quality	Utility	Trust	Acceptance & int.
Graph	280	4.06	4.30	4.11	4.05	3.94
Raw code	64	4.14	4.52	4.12	4.16	4.00
$b_{\text{raw}} = 0.07$, CI95% $[-0.02, 0.20]$; repository-bootstrap interval						

block (85 students, $\chi^2(3) = 23.22$). These response exclusions are sensitivity checks, not evidence that flagged responses are invalid.

In the intercept-only model, most AAI variance lay between students: the intraclass correlation of the agreement index was .77 (also .77 without straight-lined responses), indicating substantial clustering by student. This variance decomposition does not identify why configuration differences are small.

5.4 RQ3: Raw-Code versus Graph-Based Questions

Mean AAI was 4.14 for raw-code and 4.06 for graph-based questions (Table 5). The student fixed-effects model with configuration adjustment estimated $b = 0.07$ (95% repository-bootstrap CI $[-0.02, 0.20]$). Among the 26 students exposed to both types, pooled means were 3.99 and 4.01, respectively; these are descriptive rather than a balanced paired estimate. Perceived quality averaged 4.52 versus 4.30, and transparency 4.59 versus 4.22. The routing rule ties context to method count, and the same context type feeds the evaluator. These estimates cannot separate complexity, question generation and feedback context; the interval crossing zero does not establish equivalence.

5.5 Students' Free-Text Replies

The export includes 160 optional free-text replies (46.5% of instances). We treat them as illustrations, without independently coded prevalence estimates. Replies included proposed revisions, questions about error codes and check ordering, objections to assumed

request types, and a complaint that a single feedback block was difficult to read. These student reports have not been verified against code or observed revisions, so they establish neither technical errors nor feedback uptake.

6 Discussion

Acceptance, motivation and reported use. Endorsement of accuracy and usefulness exceeded endorsement of motivation and future use. Utility was positively associated with acceptance and intention within students; trust was associated with acceptance, consistent with technology-acceptance research [5,33]. Motivation was associated with willingness when separated from psychological safety. These self-reports do not establish actual revision or correctness. Specific objections in replies illustrate the need to verify appropriate reliance independently [23].

Configuration bundles motivate controlled tests. Small AAI differences coexisted with an actionability gap of about 0.37 points between Sandwich and Explainable. The configurations designed to include next steps had higher mean actionability, but content, scoring, length and model orchestration also differed. An experiment manipulating next steps independently is needed. Researching produced more text without a detected AAI advantage over Sandwich; efficiency claims require usage logs. The high student ICC supports accounting for individual differences in comparisons.

Response patterns and measurement. Each student supplied sixty ratings; 18.6% of response sets were straight-lined. Response burden and reverse wording warrant scrutiny [21,32]. A shorter instrument, attention checks and behavioural probes merit evaluation; straight-lining alone does not establish inattention.

Threats to validity. Internal validity. Assignment within students limits stable individual differences, but each configuration evaluates a different answer. Question difficulty, order and carryover remain possible sources of variation; the comparison cannot isolate individual content features. Exported Researching model roles are not independently corroborated by provider logs. *Construct validity.* The study-specific instrument lacks established factorial validity. AAI spans heterogeneous domains, exclusion of Rejection was data-informed, and the affect pair has low reliability. Straight-lining does not establish inattention. Self-reports collected together may share common-method variance [29]. No observed revision, learning outcome or expert correctness measure was collected, and free-text observations are illustrative rather than validated thematic prevalence estimates. *External validity.* Classroom recruitment at one university, preparation for a defence panel, and novelty may shape responses. RQ3 is observational and confounded by method count and the evaluator's context. *Statistical conclusion validity.* Student blocks are used in rank comparisons, and repository resampling preserves shared-student and shared-question dependence for the added estimates. Rank tests and student-intercept models do not model all of this dependence. Predictor regressions isolate within-student variation, but do not establish causal effects or adjust for every possible time-varying confounder. High ratings may limit discrimination. Mixed models assume interval scaling and serve as sensitivity analyses; the AAI omnibus Friedman and mixed-model tests

differ at the .05 threshold. The ±0.25 equivalence margin is pragmatic, exploratory and specific to AAI. The fifteen exploratory regression intervals are unadjusted for multiplicity, and only 28 repository clusters are available. Bootstrap coverage with this number of clusters is approximate; neither classroom sampling nor repository resampling makes the sample representative.

7 Conclusion

PACE connects repository analysis, scenario questions and structured feedback in a classroom comprehension workflow. Across 344 instances from 86 students, perceived feedback ratings were positive, but endorsement of future use and motivation was lower than endorsement of accuracy and usefulness. Small AAI differences across four assigned configurations coexisted with larger actionability differences. The bundled configurations do not isolate next steps or model effects, and the context comparison was inconclusive. Findings concern perceptions rather than correctness or learning.

Future work should audit execution logs, validate the instrument, measure revisions and expert correctness, and manipulate next steps and context independently.

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Research ethics and data governance Author confirmation required before submission: classroom recruitment is described in Section 4. Voluntariness, any relationship to grading, consent for research use, institutional review or exemption, and safeguards for student identifiers and repository content remain to be documented. No approval or consent procedure is asserted without supporting records.

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